STAT 312: Exam 02 Review

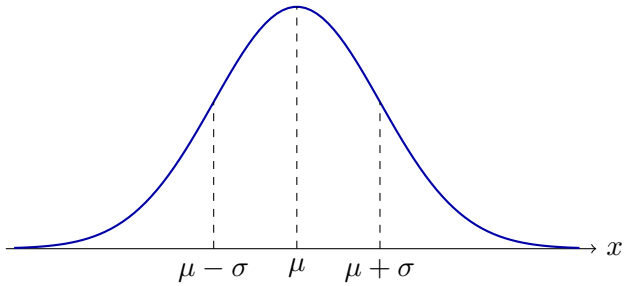
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Unit 06: Normal Distributions & Central Limit Theorem

Throughout this review, $N(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2. Its standard deviation is $\sigma > 0$.

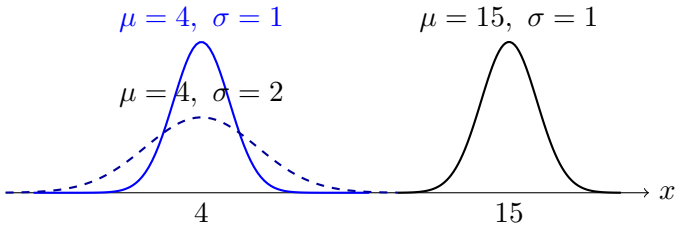
The Normal Distribution

A normal distribution has a bell-shaped density that is symmetric about μ. The total area under the curve is 1. The mean determines its center, and the standard deviation determines its spread.



Changing the Mean and Standard Deviation

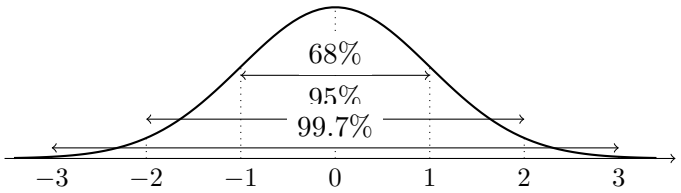
Changing μ shifts the curve horizontally. A larger σ produces a wider, lower curve, while a smaller σ produces a narrower, taller curve. Each curve still has total area 1.



The Empirical Rule

For a normal population, approximately

$$P(\mu - \sigma < X < \mu + \sigma) \approx 0.68,$$
$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95,$$
$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997.$$



The Standard Normal Distribution

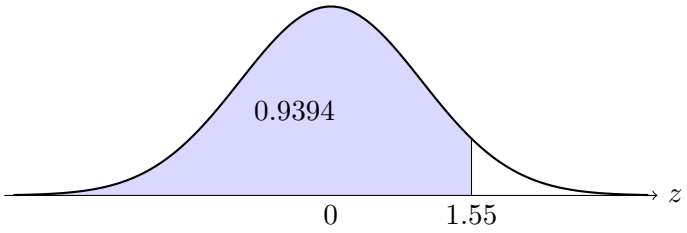
If $X \sim N(\mu, \sigma^2)$, then

$$\boxed{Z = \frac{X - \mu}{\sigma} \sim N(0, 1)}.$$

Reading a cumulative standard normal table. The table gives the area to the left of z. Use the row for the units and tenths digits and the column for the hundredths digit. Check the heading of any table you use, since some tables report different areas.

Example. To find $P(Z < 1.55)$, use row 1.5 and column 0.05:

$$\boxed{P(Z < 1.55) = \Phi(1.55) = 0.9394}.$$



Related practice from the class handout: Unit 06 handout, Examples 1–6.

Random Samples, Statistics, and Estimators

Random sample. $X_1, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables from the population.
Statistic. A function of the observed sample that does not depend on unknown population parameters.
Sampling distribution. The probability distribution of a statistic over repeated random samples of the same size.
Estimator. A statistic used to estimate a population parameter.
Estimate. The numerical value of an estimator for a particular observed sample.

	Mean	Variance	Standard deviation
Population parameter	μ	σ^2	σ
Sample statistic	$\overline{X}$	S^2	S
Observed value	$\overline{x}$	s^2	s

Different random samples generally give different sample means. Thus $\overline{X}$ is a random variable with its own mean, variance, and standard deviation.

Mean and Variance of the Sample Mean

Suppose $X_1, \ldots, X_n$ are i.i.d., with $E(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Define

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

By linearity of expectation,

$$E(\overline{X}) = \frac{1}{n} \sum_{i=1}^{n} E(X_i) = \frac{n\mu}{n} = \boxed{\mu}.$$

Using independence to add variances,

$$\mathrm{Var}(\overline{X}) = \frac{1}{n^2} \mathrm{Var}\left(\sum_{i=1}^{n} X_i \right) = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(X_i) = \boxed{\frac{\sigma^2}{n}}.$$

Therefore, the standard deviation of $\overline{X}$, called its **standard error**, is

$$\boxed{\mathrm{SD}(\overline{X}) = \frac{\sigma}{\sqrt{n}}}.$$

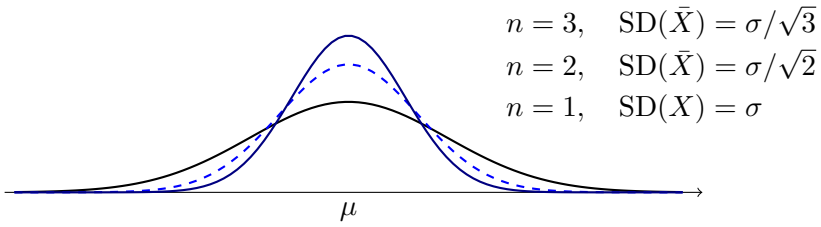
These mean and variance formulas do not require a normal population. Independence is used in the variance derivation.

Sampling Distributions and the CLT

Case 1. Normal population. If $X_1, \ldots, X_n$ are an i.i.d. sample from $N(\mu, \sigma^2)$, then for every sample size n,

$$\boxed{\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)}.$$

This is an exact result. All sampling distributions have the same center μ, and their spread decreases as n increases.



For a normal population, standardize a sample mean using

$$\boxed{Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}}.$$

A quantile of the sampling distribution is $\overline{x}_p = \mu + z_p \sigma/\sqrt{n}$.

Case 2. Population need not be normal. The central limit theorem states that, for an i.i.d. sample with finite mean μ and finite positive variance σ^2, the standardized sample mean approaches a standard normal distribution as n grows:

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

For sufficiently large n, we therefore use the approximation

$$\overline{X} \;\dot{\sim}\; N\left(\mu, \frac{\sigma^2}{n}\right).$$

How large is large enough? The guideline $n \geq 30$ is a rule of thumb. The needed sample size depends on population skewness and tail behavior. Strong skewness or heavy tails can require a substantially larger sample.

Choosing the denominator

Quantity	Standard deviation	Standardized value
X	σ	$(x - \mu)/\sigma$
$\overline{X}$	$\sigma/\sqrt{n}$	$(\overline{x} - \mu)/(\sigma/\sqrt{n})$

Related practice: Unit 06 handout, Example 7 and Example 8.

Unit B: Point Estimate

Unbiased Estimation of the Population Variance

An estimator $\widehat{\theta}$ is **unbiased** for θ if $E(\widehat{\theta}) = \theta$ for every parameter value under consideration. Since $E(\overline{X}) = \mu$, the sample mean is unbiased for μ.

Now let $X_1, \ldots, X_n$ be i.i.d. with $n \geq 2$, $E(X_i) = \mu$, and $\text{Var}(X_i) = \sigma^2 < \infty$. Define

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2.$$

We will show that $E(S^2) = \sigma^2$.

Step 1. Expand the sum of squared deviations. Using $\sum_i X_i = n\overline{X}$,

$$\begin{aligned} \sum_{i=1}^{n} (X_i - \overline{X})^2 &= \sum_{i=1}^{n} X_i^2 - 2\overline{X} \sum_{i=1}^{n} X_i + n\overline{X}^2 \\ &= \sum_{i=1}^{n} X_i^2 - 2n\overline{X}^2 + n\overline{X}^2 \\ &= \sum_{i=1}^{n} X_i^2 - n\overline{X}^2. \end{aligned}$$

Step 2. Take expectations.

$$E(S^2) = \frac{1}{n-1} \left[\sum_{i=1}^{n} E(X_i^2) - nE(\overline{X}^2) \right].$$

Step 3. Use the second-moment identity. For a random variable Y with finite variance,

$$E(Y^2) = \text{Var}(Y) + [E(Y)]^2.$$

Consequently,

$$\sum_{i=1}^{n} E(X_i^2) = n(\sigma^2 + \mu^2), \qquad E(\overline{X}^2) = \frac{\sigma^2}{n} + \mu^2.$$

Step 4. Substitute and simplify.

$$\begin{aligned} E(S^2) &= \frac{1}{n-1} \left[n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2 \right) \right] \\ &= \frac{n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2}{n-1} \\ &= \frac{(n-1)\sigma^2}{n-1} = \boxed{\sigma^2}. \end{aligned}$$

Thus S^2 is an unbiased estimator of σ^2. This proof does not require population normality.

Why Divide by $n-1$?

The previous calculation shows

$$E\left[\sum_{i=1}^{n}(X_i-\overline{X})^2\right]=(n-1)\sigma^2.$$

Dividing by $n-1$ therefore produces an unbiased estimator of σ^2.

If instead we define

$$\widetilde{S}^2=\frac{1}{n}\sum_{i=1}^{n}(X_i-\overline{X})^2=\frac{n-1}{n}S^2,$$

then

$$E(\widetilde{S}^2)=\frac{n-1}{n}\sigma^2, \qquad \boxed{\text{Bias}(\widetilde{S}^2)=-\frac{\sigma^2}{n}}.$$

For $\sigma^2>0$, this estimator systematically underestimates the variance on average.

Degrees of freedom. The deviations satisfy

$$\sum_{i=1}^{n}(X_i-\overline{X})=\sum_{i=1}^{n}X_i-n\overline{X}=0.$$

Once any $n-1$ deviations are known, the remaining deviation is determined by this constraint. There are $n-1$ degrees of freedom in the deviations.

Bias and Mean Squared Error

The bias of an estimator $\widehat{\theta}$ is

$$\text{Bias}(\widehat{\theta})=E(\widehat{\theta})-\theta.$$

Its mean squared error measures its expected squared distance from the parameter:

$$\text{MSE}(\widehat{\theta})=E\left[(\widehat{\theta}-\theta)^2\right].$$

When the estimator has finite second moment,

$$\boxed{\text{MSE}(\widehat{\theta})=\text{Var}(\widehat{\theta})+[\text{Bias}(\widehat{\theta})]^2}.$$

Unit 07: Statistical Inference and Confidence Intervals

Statistical inference uses information from a random sample to draw conclusions about a population.

- **Point estimation.** Use a single statistic to estimate a population parameter. For example, use $\overline{X}$ to estimate μ.
- **Interval estimation.** Construct an interval that describes the uncertainty associated with an estimate.
- **Hypothesis testing.** Evaluate evidence concerning a claim about a population parameter, such as $H_0 : \mu = 8$.

Many confidence intervals have the form

$$\boxed{\text{point estimate } \pm \text{ margin of error}}.$$

The margin of error often equals

$$\text{critical value} \times \text{standard error}.$$

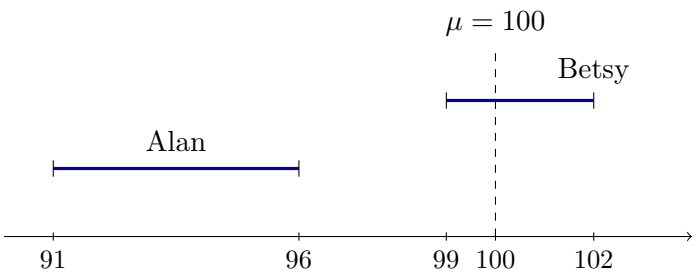
Interpreting a Confidence Level

A 95% confidence procedure captures the true population parameter in approximately 95% of intervals constructed over repeated random samples, provided its assumptions hold.

The population parameter is fixed. The interval varies from sample to sample.

Example. Suppose the true population mean is $\mu = 100$. Two samples produce the following 95% confidence intervals:

Sample	Confidence interval	Contains $\mu = 100$?
Alan's sample	$(91, 96)$	No
Betsy's sample	$(99, 102)$	Yes



Both intervals were constructed using a 95% procedure. A particular interval either contains μ or does not. In practice, μ is unknown, so we usually cannot determine whether the observed interval captured it.

Related practice: Unit 07 handout, Examples 1–3, 6, 7, 8.

From the Normal Distribution to Student's t Distribution

When σ is known and the sample mean is normally distributed,

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

When σ is unknown, estimate it using the sample standard deviation S. This gives

$$\boxed{T = \frac{\overline{X} - \mu}{S/\sqrt{n}}}.$$

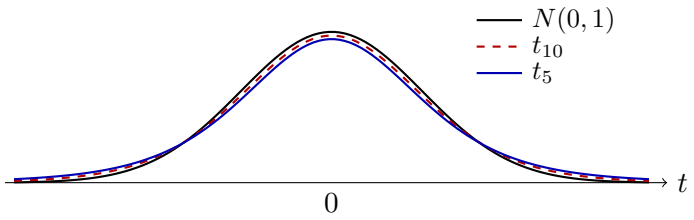
If $X_1, \ldots, X_n$ are an independent random sample from a normal population, then

$$\boxed{T \sim t_{n-1}}.$$

Replacing σ with S introduces additional uncertainty. The t distribution accounts for that uncertainty.

Properties of the t Distribution

- The density is symmetric about zero.
- The total area under the density curve is 1.
- For finite degrees of freedom, the density has a lower peak and heavier tails than the standard normal density.
- Its shape depends on its degrees of freedom, denoted by ν.
- As ν increases, the t distribution approaches the standard normal distribution.



Degrees of Freedom

Degrees of freedom describe the number of independent pieces of information available after accounting for constraints or estimated quantities.

For the one-sample t procedure,

$$\boxed{\nu = n - 1}.$$

The sample deviations satisfy

$$\sum_{i=1}^{n} (X_i - \overline{X}) = 0.$$

Once $n - 1$ deviations are known, the final deviation is determined.

For this procedure, the degrees of freedom are positive integers when $n \geq 2$. More generally, some statistical procedures use noninteger degrees of freedom.

Reading a Student's t Table

See class notes and lecture notes for examples.

A Confidence Interval for a Population Mean

Known Population Standard Deviation

For an independent random sample from a normal population with known σ, a $(1-\alpha)100\%$ confidence interval for μ is

$$\boxed{\overline{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}.$$

For a nonnormal population, this interval can be used approximately when the normal approximation for $\overline{X}$ is adequate (e.g., $n \geq 30$).

Unknown Population Standard Deviation

For an independent random sample, if σ is unknown and $\overline{X}$ is normally (or approximately normally) distributed, a $(1-\alpha)100\%$ confidence interval for μ is

$$\boxed{\overline{X} \pm t_{\alpha/2,\,(n-1)}\frac{S}{\sqrt{n}}}.$$

For a nonnormal population, this interval can be used approximately when the normal approximation for $\overline{X}$ is adequate (e.g., $n \geq 30$).

Related practice: Unit 07 handout, Example 9-11.

Estimating a Population Proportion

Let p be the population proportion possessing a particular characteristic, where $0 \leq p \leq 1$.

For a sample of size n, let X be the number of observations possessing that characteristic. The sample proportion is

$$\boxed{\widehat{p} = \frac{X}{n}}.$$

Under independent Bernoulli trials with a common success probability p,

$$X \sim \text{Bin}(n, p),$$

with

$$E(X) = np, \qquad \text{Var}(X) = np(1-p).$$

Mean of the Sample Proportion

By linearity of expectation,

$$\begin{aligned} E(\widehat{p}) &= E\left(\frac{X}{n}\right) \\ &= \frac{1}{n}E(X) \\ &= \frac{np}{n} \\ &= \boxed{p}. \end{aligned}$$

Thus $\widehat{p}$ is an unbiased estimator of p.

Variance and Standard Deviation

$$\begin{aligned} \mathrm{Var}(\widehat{p}) &= \mathrm{Var}\left(\frac{X}{n}\right) \\ &= \frac{1}{n^2}\,\mathrm{Var}(X) \\ &= \frac{np(1-p)}{n^2} \\ &= \boxed{\frac{p(1-p)}{n}}. \end{aligned}$$

Therefore,

$$\boxed{\mathrm{SD}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}}.$$

Because p is unknown, we estimate this standard error using the observed sample proportion:

$$\boxed{\widehat{\mathrm{SE}}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}}.$$

A Confidence Interval for a Population Proportion

The usual normal-approximation, or Wald, confidence interval for p is

$$\boxed{\widehat{p} \pm z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}}.$$

Its margin of error is

$$E = z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

Conditions

1. **Random sampling.** The data should come from a random sample or an appropriate random sampling model.
2. **Independence.** Observations should be independent.
3. **Enough successes and failures.** Using the threshold in these class notes, check

$$\boxed{n\widehat{p} \geq 15 \qquad \text{and} \qquad n(1-\widehat{p}) \geq 15}.$$

   These are the observed numbers of successes and failures.

Common Standard Normal Critical Values

The notation $z_{\alpha/2}$ denotes a positive value with area $\alpha/2$ to its right.

Confidence level	α	$\alpha/2$	$z_{\alpha/2}$
90%	0.10	0.05	1.645
95%	0.05	0.025	1.960
99%	0.01	0.005	2.576

Related practice: Unit 07 handout, Example 12.

Unit 08: Hypothesis Testing

A hypothesis test uses sample data to evaluate a claim about an unknown population parameter.

- The **null hypothesis**, H_0, specifies the claim used to determine the reference distribution of the test statistic.
- The **alternative hypothesis**, H_a, states the competing claim for which we seek evidence.
- Formulate the hypotheses and choose the significance level before examining the data for the test.

Example. A candy company claims that its candies have mean mass 1 gram. We suspect that the mean is smaller:

$$H_0 : \mu = 1, \qquad H_a : \mu < 1.$$

For a mean, the three standard alternatives are

Alternative	Type of test
$\mu < \mu_0$	Left-tailed
$\mu > \mu_0$	Right-tailed
$\mu \neq \mu_0$	Two-tailed

Hypotheses concern population parameters, such as μ or p. Sample statistics, such as $\overline{x}$ or $\widehat{p}$, supply the evidence used to evaluate those hypotheses.

The Two Possible Decisions

- **Reject H_0.** The data provide sufficient evidence against H_0 at the chosen significance level.
- **Fail to reject H_0.** The data provide insufficient evidence against H_0 at that level.

Failing to reject H_0 does not establish that it is true. The classroom courtroom analogy illustrates this distinction: insufficient evidence for a guilty verdict does not establish innocence.

Type I Error, Type II Error, and Power

	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

The **significance level**, also called the **level of significance**, is

$$\begin{aligned} \alpha &= P(\text{Type I error}) \\ &= P(\text{reject } H_0 \mid H_0 \text{ is true}). \end{aligned}$$

For a specified parameter value under the alternative, the probability of a Type II error is

$$\begin{aligned} \beta &= P(\text{Type II error}) \\ &= P(\text{fail to reject } H_0 \mid H_0 \text{ is false}). \end{aligned}$$

The **power of the test** is

$$\boxed{\begin{aligned}1-\beta &= P(\text{reject } H_0 \mid H_0 \text{ is false})\\ &= \text{power of the test.}\end{aligned}}$$

The significance level α controls the Type I error probability. For an exact continuous test at level α, this probability is α under the stated null model. Power and β depend on the particular true parameter value, sample size, and test procedure.

The One-Sample z Test for a Mean

For $H_0 : \mu = \mu_0$ with known population standard deviation σ,

$$\boxed{z = \frac{\overline{x} - \mu_0}{\sigma/\sqrt{n}}}.$$

The test is exact for an independent random sample from a normal population. For a nonnormal population, it is approximate when the normal approximation for the sample mean is adequate.

Write z_α for the positive critical value satisfying $P(Z > z_\alpha) = \alpha$, for the usual choices $0 < \alpha < 0.5$.

Alternative	Rejection region
$H_a : \mu < \mu_0$	$z < -z_\alpha$
$H_a : \mu > \mu_0$	$z > z_\alpha$
$H_a : \mu \neq \mu_0$	$\lvert z\rvert > z_{\alpha/2}$

Related practice: Unit 08 handout, Example 2.

Example A. A Two-Tailed Test (Class Notes)

Suppose

$$H_0 : \mu = 50, \quad H_a : \mu \neq 50, \quad \alpha = 0.01, \quad \overline{x} = 53, \quad n = 64, \quad \sigma = 16.$$

Then

$$z_{\text{obs}} = \frac{53 - 50}{16/\sqrt{64}} = \boxed{1.5}.$$

The critical values are approximately ± 2.576. Since $|1.5| < 2.576$, **fail to reject** H_0. There is insufficient evidence at the 1% level that μ differs from 50.

Example B. A Right-Tailed Test (Class Notes)

Suppose

$$H_0 : \mu = 70, \quad H_a : \mu > 70, \quad \alpha = 0.09, \quad \overline{x} = 74, \quad n = 36, \quad \sigma = 12.$$

Then

$$z_{\text{obs}} = \frac{74 - 70}{12/\sqrt{36}} = \boxed{2}.$$

The upper-tail critical value is $z_{0.09} \approx 1.341$. Since $2 > 1.341$, **reject** H_0. The data provide evidence at the 9% level that $\mu > 70$.

For these same data, the upper-tail probability is approximately 0.0228. Thus the result is significant at levels 0.09 and 0.05, but not at level 0.01. These comparisons illustrate how the decision depends on the level; the level for the actual test should be chosen in advance.

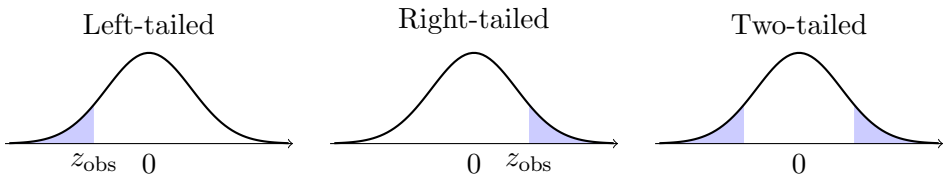
Understanding and Computing a p-Value

A p-value is the probability, assuming H_0 is true, of obtaining a test statistic (in other words, observing the data) as extreme or more extreme than the one we obtained.

A smaller p-value indicates stronger evidence against H_0. It is not the probability that H_0 is true.

Let Φ be the standard normal cumulative distribution function. For the one-sample z test,

Alternative	p-value
$\mu < \mu_0$	$\Phi(z_{\text{obs}})$
$\mu > \mu_0$	$1 - \Phi(z_{\text{obs}})$
$\mu \neq \mu_0$	$2[1 - \Phi(\lvert z_{\text{obs}} \rvert)]$



The shaded region represents the p-value. For the symmetric two-tailed test, each shaded tail contributes half the p-value.

Decision Rule

p-value $\leq \alpha$	$\Longrightarrow$	reject H_0,
p-value $> \alpha$	$\Longrightarrow$	fail to reject H_0.

For Example A,

$$p\text{-value} = 2P(Z > 1.5) \approx 2(0.0668) = \boxed{0.1336}.$$

Because $0.1336 > 0.01$, the p-value method gives the same decision as the critical-value method.

A Practical Testing Checklist

1. Define the parameter and state H_0 and H_a.
2. Choose α and check the assumptions.
3. Compute the test statistic using the null parameter value.
4. Find the p-value or compare with the rejection region.
5. State the decision and a conclusion in the original context.

Worked Example: A Left-Tailed z Test

Consider the milk-container example from the class notes. Let μ be the population mean amount in a container, using the same measurement units for all quantities. Suppose

$$H_0 : \mu = 1, \qquad H_a : \mu < 1,$$

with

$$\overline{x} = 0.94, \qquad \sigma = 0.19, \qquad n = 50.$$

Assume a random independent sample and an adequate normal approximation for the sample mean. The test statistic is

$$z = \frac{0.94 - 1}{0.19/\sqrt{50}} \approx \boxed{-2.233}.$$

The alternative is left-tailed, so

$$p\text{-value} = P(Z \leq -2.233) \approx \boxed{0.0128}.$$

Using a table with $z = -2.23$ gives 0.0129.

Interpretation. If the population mean really equals 1, approximately 1.28% of random samples of size 50 would produce a sample mean of 0.94 or less under this model.

At $\alpha = 0.10$, reject H_0 because $0.0128 < 0.10$. The data provide evidence that the population mean amount is less than 1.

The One-Sample t Test for a Mean

When σ is unknown, use the sample standard deviation s:

$$\boxed{t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}}}, \qquad \nu = n - 1.$$

The reference distribution is exactly t_{n-1} for independent observations from a normal population. For other populations, its use is approximate and depends on sample size, skewness, and outliers.

Let $t_{\alpha,\nu}$ denote the upper-tail critical value satisfying $P(T_\nu > t_{\alpha,\nu}) = \alpha$.

Alternative	Rejection region
$\mu < \mu_0$	$t < -t_{\alpha,\nu}$
$\mu > \mu_0$	$t > t_{\alpha,\nu}$
$\mu \neq \mu_0$	$\lvert t \rvert > t_{\alpha/2,\nu}$

The p-value is the corresponding left-tail, right-tail, or combined two-tail area under the t_ν distribution.

Worked Example: A Left-Tailed t Test (Class Notes)

Let μ be the population mean cholesterol level among 70-year-old American men, on the measurement scale used in the class example. Assume an independent random sample from a normal population. Suppose

$$H_0 : \mu = 45.5, \qquad H_a : \mu < 45.5,$$

with

$$n = 19, \qquad \overline{x} = 44.2, \qquad s = 3.8, \qquad \alpha = 0.05.$$

Then

$$t_{\text{obs}} = \frac{44.2 - 45.5}{3.8/\sqrt{19}} \approx \boxed{-1.491}, \qquad \nu = 18.$$

The rejection region is

$$t_{\text{obs}} < -t_{0.05,18} \approx -1.734.$$

Since $-1.491 > -1.734$, **fail to reject** H_0. There is insufficient evidence at the 5% level that the population mean is below 45.5.

The table gives $t_{0.10,18} \approx 1.330$ and $t_{0.05,18} \approx 1.734$. Therefore,

$$\boxed{0.05 < p\text{-value} < 0.10}.$$

A calculator gives p-value ≈ 0.0766.

Bounding p-Values Using a t Table (Class Notes)

Exercise 1. Right-Tailed Test

Suppose $n = 26$ and $t_{\text{obs}} = 2.1$. Then $\nu = 25$. The table gives

$$t_{0.025,25} \approx 2.060, \qquad t_{0.01,25} \approx 2.485.$$

Because $2.060 < 2.1 < 2.485$, the right-tail probability satisfies

$$\boxed{0.01 < p\text{-value} < 0.025}.$$

The numerical value is approximately 0.0230.

Exercise 2. Two-Tailed Test

Suppose $n = 21$ and $t_{\text{obs}} = 2$. Then $\nu = 20$. The table gives

$$t_{0.05,20} \approx 1.725, \qquad t_{0.025,20} \approx 2.086.$$

Since $1.725 < 2 < 2.086$, one tail has area between 0.025 and 0.05. Double these bounds for the two-tailed test:

$$\boxed{0.05 < p\text{-value} < 0.10}.$$

The numerical value is approximately 0.0593.

A Hypothesis Test for a Population Proportion

Let p be the population proportion with the characteristic of interest. For x successes in a sample of size n, define $\widehat{p} = x/n$. Test

$$H_0 : p = p_0$$

against one of

$$H_a : p < p_0, \qquad H_a : p > p_0, \qquad H_a : p \neq p_0.$$

Under H_0, the standard error used for the test is

$$\mathrm{SE}_0(\widehat{p}) = \sqrt{\frac{p_0(1-p_0)}{n}}.$$

Thus the one-proportion z statistic is

$$\boxed{z = \frac{\widehat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}}.$$

The denominator uses the null proportion p_0. For the confidence interval in Part 2, the estimated standard error instead uses $\widehat{p}$.

Decision

Use the standard normal rejection regions and p-value formulas, with the tail selected by H_a. Reject H_0 if the p-value is at most α.

Related practice: Unit 08 handout, Example 3.